

Xingyun Mou

888 South Hope Street ♦ xingyunm@usc.edu ♦ (+1)9496683386

EDUCATION

University of Southern California

*Candidate for Bachelor of Science in Applied and Computational Mathematics
Minor in Artificial Intelligence Applications*

Los Angeles, CA
Jan 2024 - Present

- **Cumulative GPA: 3.83/4.0**
- **Honor:** 2024 USC Dornsife Dean's List, 2025 USC Dornsife Dean's List
- **Relevant Courses:** Artificial Intelligence, Object Oriented Programming, Discrete Methods in Computer Science, Machine Learning, Probability Theory

University of California Irvine

*Candidate for Bachelor of Arts in Environmental Science and Policy
Double major in Urban Studies*

Irvine, CA
Sep 2022 - Dec 2023

- **Cumulative GPA: 4.0/4.0**
- **Honor:** 2022 Dean's Honor List, 2023 Dean's Honor List
- **Relevant Courses:** Linear Algebra, Basic Statistics, Calculus

EXPERIENCE

Shanghai AI Laboratory

Research Intern

Shanghai, China
May 2025 - Dec 2025

- Conducted supervised fine-tuning on DeepSeek-R1-Distill-Qwen-7B with explicit Chain-of-Thought supervision to enhance multi-step mathematical reasoning and improve faithfulness between intermediate reasoning and final answers
- Designed and evaluated a two-stage SFT framework combining Chain-of-Thought format alignment with hard-example refinement demonstrating clear improvements in reasoning depth logical coherence and solution stability over the base model
- Built and preprocessed large-scale datasets for model pretraining, including data cleaning, normalization, and formatting to support stable training runs
- Investigated Lean4-based formal verification in improving LLM mathematical reasoning capabilities

Mathematics Tutor

Tutor

Los Angeles, CA
Sep 2024 - Dec 2024

- Assisted students during weekly sessions by answering questions and emphasizing key solution steps
- Explained core Calculus II concepts using structured problem-solving approaches

Alex Guenther's Lab

Researcher

Irvine, CA
Nov 2022 - Nov 2023

- Processed and analyzed large-scale datasets (satellite imagery, geospatial layers, time-series data) to extract structured features for modeling
- Used QGIS and remote sensing data to perform 30-year land cover change analysis
- Identified nonlinear relationships and regional patterns in high-dimensional data using exploratory data analysis and correlation analysis

Orange County Sustainability Decathlon

Leader of Marketing-Communications-Design Team

Irvine, CA
Jan 2023 - May 2023

- Optimized 3D structural and heat-transfer models via finite element and data-driven analysis
- Analyzed survey datasets to derive quantitative requirements for affordable housing systems
- Communicated technical insights with nonprofit stakeholders to align objectives and secure funding

SKILLS AND INTERESTS

Language: Fluent - English and Native - Mandarin

Technical Skills: C++, Python, R, Git, Linux, MATLAB, QGIS, EXCEL, LaTeX